

General Features

The i211 is an intelligent high precision analog interface module for the ionia™ system. The module has 12 analog inputs grouped into four galvanically isolated sections, each with its own reference ground.

Each group can be individually selected as voltage, resistive or current measurement inputs. All channels are factory-calibrated and have a maximum resolution of 24 bits. The sampling rate is configurable up to a maximum of 25 Hz.

All configuration registers and the input values are accessible by a fast and secure point to point connection from the central unit. The on-board controller supports independent

digital low-pass filters and other calculation functions, that can be customized by the application developers.

A built-in supervisory mechanism called «Condition Monitoring» can detect, record and report any off-limit usage, such as overvoltage, overcurrent, supply interruption, overtemperature, etc. as well as selected user-defined values.

Configuring the i211 is done easily and time saving by using one of the five methods defined by IEC 61131.

The i211 is designed for harsh rolling stock environments and is fully compliant to EN 50155.

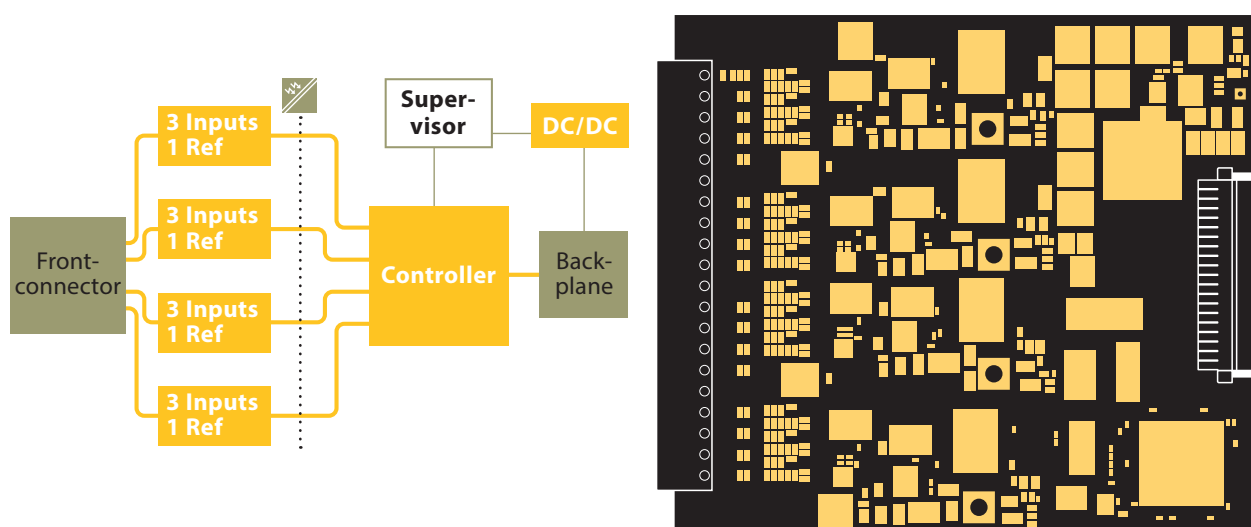
Key Benefits

- Configurable digital filters and functions
- Status LEDs on each channel for easy diagnostics
- Galvanically separated groups for independent measurements
- User configurable sampling rates
- «Condition Monitoring» sends detailed diagnostics to the central unit
- IEC 61131 for fast and easy programming
- No capacitor replacement necessary
- Complies to EN 50155, EN 50121, IEC 61373

Application Examples

- Temperature measurement with PT1000/PT100
- Humidity sensor
- Voltage monitoring
- Tachometer measurement
- Ambient light measurement
- Current measurements
- Battery monitoring
- Resistive measurements

i211 Hardware Architecture



Life Cycle Cost

The i211 has been strictly designed without electrolytic capacitors to avoid service costs for changing capacitors. The use of an FPGA with included duagon soft-IP minimizes

the risk and cost in case of component obsolescence. Any firmware upgrade can be done via the central unit, without the need of removing the device.

Technical Data

Input Channels	– 12 input channels divided into four groups – Each group has there own reference ground – 1000 Vac galvanic isolation between groups – Inputs selectable as – Voltage input: 0 to 10 Vdc with 50 kΩ input resistance – Current loop: 0 to 20 mA, input impedance: 120 Ω – Resistive: 0 kΩ to 10 kΩ, or 2 wire PT100/PT1000 – 3 or 4 wire PT100/PT1000 – Absolute precision: (–40 °C to +85 °C) 0.2% of full scale – Effective measurement resolution: normal 24 bits, fast 16 bits – Max conversion rate of each specific channel: 25 Hz (software selectable)	Diagnostic/ Service	– Configurable diagnostic LEDs for each channel indicating the status of the channel or a user-defined functionality – LEDs for visual status information (power on, running, communicating, failure, I/O, ...) – Diagnostic information readable through the central unit's service interfaces (see i101/i102/i103 for further information about these possibilities)
Configuration	– Programmable by the central unit using IEC 61131 compliant executor and IDE supporting LD, FBD, ST, IL and SFC	Power Consumption	– $P_{max} < 2\text{ W}$ – Supports sleep mode – Fully powered via backplane (no external power supply needed) – Interruption Class S2 with ceramic capacitors
Security/ Watchdog	– External supervisor for voltage, current and temperature monitoring → «Condition Monitoring» – Communication Link with watchdog supervision	Operating Conditions	– Ambient temperature: –40 to +85 °C (EN 50155, Class TX) – Relative humidity: 75%, max 95% for 30 days per year (conformal coating) EN 60068 – Shock and vibration: According to IEC 61373 category 1, class B – EMI: According to EN 50121 and EN 50155
Backplane Communication	– 100 Mbit duagon serial star slave – Secure communication protocol – Configurable status scan cycle time minimum: 1 ms	Dimensions/ Weight	– 3TE × 92 × 86 mm, fitting the ionia™ housing – 120 g
		Environment	– Fully compliant with RoHS and REACH

d-006355-013034

Order Code and HW Options

Product Type: **i211** - - - -

EXAMPLE:
i014-V-V-C-P

Input Type G1		Input Type G2		Input Type G3		Input Type G4	
Voltage	V	Voltage	V	Voltage	V	Voltage	V
Current	C	Current	C	Current	C	Current	C
Resistance	R	Resistance	R	Resistance	R	Resistance	R
PT100/PT1000, 4 wire	P	PT100/PT1000, 4 wire	P	PT100/PT1000, 4 wire	P	PT100/PT1000, 4 wire	P

Related Documents

Data Sheet i211  d-006660-nnnnnn.pdf
available at www.duagon.com

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